FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

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Project: Northstar Outfitters — E-Commerce Platform Modernization

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1. DOMAIN: INVENTORY & SYNCHRONIZATION

FR-INV-001 — Sellable Inventory Calculation

The system shall calculate the available sellable stock for online purchase using the pessimistic rule between the local inventory state and external ERP updates.

- **BR-INV-001 — Pessimistic Conflict Rule:** When a discrepancy exists between local e-commerce stock and the external ERP stock, the system must adopt the lower quantity as the authoritative Sellable Quantity. An operational alert must be automatically emitted for manual audit.
- **AC-INV-001.1 — Non-Negative Inventory:** The system shall never allow Sellable Quantity to drop below zero (0). Any incoming sync result indicating negative values (e.g., -1) must be rejected, flagged as a Data Error, and logged.
- **AC-INV-001.2 — Out-of-Stock Purchase Prevention:** When Sellable Quantity = 0, the platform must disable the "Add to Cart" and "Checkout" actions for that specific SKU across all customer channels.

FR-INV-002 — Inventory Reservation & TTL

The system shall temporarily reserve inventory when a customer submits a checkout/payment request to prevent race conditions and overselling.

- **BR-INV-002 — 15-Minute Reservation TTL:** Reserved inventory shall expire automatically after 15 minutes if the payment is not completed, returning the units to public available stock.
- **BR-INV-003 — Reservation Trigger Point:** Inventory reservation must occur strictly at the moment the customer submits the payment authorization form, not upon opening the payment page.
- **AC-INV-002.1 — Active Payment Hold Exemption:** If a payment transaction is actively being processed by a payment gateway (awaiting asynchronous response), the 15-minute TTL expiration must be suspended until a definitive gateway status is received.

- **AC-INV-002.2 — Concurrency Locking:** The inventory service must execute atomic pessimistic/optimistic database locks during stock reservation to guarantee that simultaneous checkout attempts for the last available unit do not cause over-allocation.

FR-INV-003 — Automated Inventory Compensation

The system shall execute automated financial and inventory compensation workflows when payment succeeds but inventory allocation fails.

- **BR-INV-004 — Automated Refund Preference:** If an order reaches a paid state on the gateway but inventory cannot be allocated, the system must automatically initiate a full refund rather than queuing the order for manual review.
- **AC-INV-003.1 — Automated Refund Execution:** Upon inventory allocation failure post-payment, the system shall call the payment provider's refund API, transition the order to CANCELLED_REFUNDED, and notify the customer.
- **AC-INV-003.2 — High-Priority Incident Fallback:** If the automated refund API call fails or times out during a compensation workflow, the system must immediately raise a High-Priority Operational Incident for manual intervention by Customer Service/Operations.

2. DOMAIN: ORDERS & FULFILLMENT

FR-ORD-001 — Order Creation & State Machine

The system shall manage the complete order lifecycle using a deterministic, event-driven state machine.

- **BR-ORD-001 — Price Immutability:** An order must record and lock the product prices, taxes, and discounts at the exact moment of purchase. Subsequent catalog price modifications shall not alter existing order totals.
- **BR-ORD-002 — Authoritative Webhook Confirmation:** An order shall transition to the PAID state exclusively upon receipt and verification of a signed asynchronous webhook/IPN from the payment provider. Synchronous checkout responses shall only be used for UI feedback.
- **AC-ORD-001.1 — Order Lifecycle Transitions:** The system shall enforce the following strict status transitions:

PENDING_PAYMENT => PAID => PROCESSING => SHIPPED => DELIVERED.

- **AC-ORD-001.2 — Asynchronous Browser Disconnect Recovery:** If a customer's browser disconnects or times out after payment completion, the backend webhook processor must transition the order to PAID, release/commit the inventory, and dispatch confirmation emails independently of the customer's browser state.

FR-ORD-002 — Item-Level Partial Cancellation

The system shall support line-item level cancellation and tracking within a single order.

- **BR-ORD-003 — Partial Fulfillment & Refund:** If an individual item within a multi-item order cannot be fulfilled, the system shall allow Operations to cancel that specific line item, adjust the order balance, and issue a partial refund for the unavailable item while fulfilling the remainder of the order.
- **AC-ORD-002.1 — Granular Line-Item Statuses:** Each order line item must independently maintain its status (e.g., Item A: FULFILLED, Item B: CANCELLED_REFUNDED).
- **AC-ORD-002.2 — Partial Capture/Refund Recalculation:** The system shall re-evaluate order totals, applicable taxes, and pro-rated promotion discounts whenever a partial cancellation occurs.

FR-ORD-003 — Multi-Location Fulfillment (Split Shipments)

The system shall support fulfilling items in a single customer order from multiple physical locations (e.g., Central Warehouse, Store A, Store B).

- **BR-ORD-004 — Single Customer Order / Multiple Tracking:** A multi-location order must present a consolidated view to the customer while generating distinct fulfillment tasks and tracking numbers for each dispatch location.
- **AC-ORD-003.1 — Fulfillment Group Generation:** The system shall partition an order into separate "Fulfillment Groups" based on item stock location.
- **AC-ORD-003.2 — Multi-Carrier Tracking Notification:** The platform shall send incremental shipping notifications to the customer as each individual Fulfillment Group is dispatched, linking multiple tracking numbers to the parent order.

3. DOMAIN: PAYMENTS & RECONCILIATION

FR-PAY-001 — Payment Idempotency & Duplicate Protection

The system shall guarantee that duplicate payment requests for the same order do not result in multiple charges.

- **BR-PAY-001 — Transaction Idempotency:** Every payment request sent to an external provider must include a unique, deterministically generated Idempotency Key (Order_ID + Attempt_UUID).
- **AC-PAY-001.1 — Double-Click Prevention:** If a customer submits a payment request multiple times (e.g., double-clicking "Pay" or retrying during a network lag), the system must intercept the duplicate request using the idempotency key and return the original transaction result without re-charging the payment method.

FR-PAY-002 — Provider Abstraction Layer

The system shall implement an abstraction gateway to normalize feature discrepancies across multiple payment providers.

- **BR-PAY-002 — Provider Capability Mapping:** The system must maintain a feature-capability matrix (e.g., supports_partial_refund, supports_async_capture) per provider and validate requested operations before execution.
- **AC-PAY-002.1 — Graceful Degradation:** If a administrative user attempts an operation (such as a native partial refund) on a payment provider that does not support it via API, the system must block the action, display a clear explanatory message, and route the workflow to an approved manual procedure.

4. DOMAIN: CANCELLATIONS, RETURNS & REFUNDS

FR-REF-001 — Customer & Operations Cancellation Rules

The system shall enforce role-based and state-dependent cancellation policies.

- **BR-REF-001 — State-Dependent Cancellation Rights:**
    - PENDING_PAYMENT / PAID: Customer can self-cancel via self-service portal.
    - PROCESSING: Customer cannot self-cancel; can only submit a Cancellation Request to Customer Service/Operations.
    - SHIPPED / DELIVERED: Cancellation is strictly prohibited; must follow the Return Process.

- **AC-REF-001.1 — Processing State Interception:** When a cancellation is requested for an order in PROCESSING, the system must verify with the fulfillment module whether physical dispatch has occurred before approving the cancellation and releasing inventory.

FR-REF-002 — Refund Execution & Shipping Fee Rules

The system shall allow authorized staff to initiate full or partial refunds directly from the admin platform according to corporate financial policies.

- **BR-REF-002 — 30-Day Return Window:** Return and refund requests must be submitted within 30 calendar days following the order's DELIVERED timestamp.
- **BR-REF-003 — Conditional Shipping Fee Refunds:** Original shipping fees shall be refunded **only** if the return/cancellation is caused by a Northstar error (out of stock, wrong item, damaged goods). Shipping fees shall **not** be refunded for standard customer-initiated returns.
- **AC-REF-002.1 — Admin Refund Audit Trail:** Every refund executed via the admin panel must record: Original Transaction Reference, Refunded Amount, Refund Reason, Initiating Admin User ID, Timestamp, and Provider Response Payload.
- **AC-REF-002.2 — Maximum Refund Constraint:** The system shall block any refund attempt where Total Refunded Amount > Originally Captured Amount.

5. DOMAIN: ADMINISTRATION & ROLE-BASED ACCESS CONTROL (RBAC)

FR-ADM-001 — Role-Based Access Control

The system shall enforce strict Role-Based Access Control (RBAC) across all administrative functions.

- **BR-ADM-001 — Least Privilege Principle:** Users shall only access administrative tools, customer PII, and financial operations explicitly granted by their assigned role.
- **AC-ADM-001.1 — Default Roles:** The system shall provide at minimum four pre-configured roles:
    - E-Commerce Administrator: Full access to Catalog, Pricing, Promotions, and System Config.

- Operations / Fulfillment: Access to Orders, Inventory Allocation, Shipments, and Dispatches.
- Customer Service: Read access to Orders/Customers; permission to initiate cancellations, track shipments, and execute refunds under defined financial thresholds.
- Management: Read-only access to executive dashboards, sales reporting, and operational KPIs.